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MEETING AGENTIC AI'S DATA QUALITY NEEDS WITH ZERO TRUST DATA QUALITY

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ABSTRACT
As software systems enter the era of agentic AI—marked by increased autonomy and complexity—traditional approaches to data quality management fall short. While the 1980s demanded defensive coding and the 2010s formalized data quality management practices, agentic AI requires a new paradigm. This paper introduces Zero Trust Data Quality (ZTDQ), a framework that applies zero trust principles from cybersecurity to data quality assessment. We argue that data quality should always be verified, never assumed, especially as systems evolve and data undergoes transformation over time.

INTRODUCTION:
THE NEW PARADIGM OF ZERO TRUST DATA QUALITY

Just as large-scale systems of the 1980's required a new defensive coding paradigm, and the complex systems of the 2010's required formal data quality management, agentic AI requires a paradigm shift to meet new data quality needs. This paper proposes Zero Trust Data Quality (ZTDQ) as the paradigm that accounts for how agentic AI increases the complexity and autonomy of software systems by orders of magnitude.[1]

Current industry best practice, as outlined by DAMA—DMBOK, instructs us to monitor data quality throughout the data lifecycle from data creation (production) to storage, integration (aggregation), and use (consumption). Historically, achieving full coverage in the aggregation and consumption layers has proved to be somewhere between impractical and infeasible.[2]

Consider the following commonplace situation as exemplary of the challenges in the consumption layer. A firm on boarded a new account in 2010 and validated its data quality in accordance with the firm's data quality management process. Since then, the firm went through two mergers, three data center migrations, a database migration, a data architecture uplift, and a cloud migration. Is the data still in the same condition it was in when it was acquired and validated?

Furthermore, the application that accesses the data went through two rewrites, new business intelligence tools emerged, machine learning gained traction, and LLM entered the fray. Is the data still fit for purpose? Does the data remain fit for purpose if it is transformed as it travels through a series of hops, causing data dynamism, and if users edit the data? As the data ages, does it remain fresh enough for purpose or it expires? Now the firm has plans to launch a new product powered by Agentic AI. Is the data fit for the next-level new purpose?

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[1] A software system can be defined as an engineered collection of interdependent software components and data, operating within a defined architecture to fulfill specific tasks, automate processes, and enable interaction between users and other systems, often at scale and with reliability constraints.
[2] Data Management Association – Data Management Body of Knowledge. Available at: https://dama.org/learning-resources/dama-data-management-body-of-knowledge-dmbok/

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Multiple Lines of Defense

A typical Chief Data Officer (CDO) solution to the aforementioned challenges is to implement second and third lines of defense to periodically validate that the data repository is fit for all current purposes. Additional lines of defense are helpful, especially from a defensive quality standpoint. Regulators appreciate firms that consider the challenges previously outlined and implement additional lines of defense to address them.[3]

Unfortunately, the efficacy of the additional defenses is questionable, particularly with regards to the offensive data quality needed to pursue an agentic AI vision. After the fact, often manual, reviews are insufficient. This is the entry point for the promise of zero trust, a concept borrowed from the cyber security domain, whose fundamental tenet is that no user or process should be trusted within a network, even if the user was previously authenticated at the perimeter. The user should be challenged to prove their identity every time they move through, or take action, within the network, regardless of whether they have previously authenticated. When applied to data quality, we can say that data quality should always be verified, never trusted.

The concept of applying zero trust principles to data quality assessment was introduced by Mohammed et al. (2021), who argued that to address data dynamism challenges, data quality should be evaluated at each hop in the ETL process and classified as either trusted data or exception data.[4] Our goal in this paper is to build upon this previous work in presenting the design principles for a new data quality paradigm – one that applies the full zero trust security concept to data quality.

We propose that data should never be considered permanently trusted - its trustworthiness should be valid only for a specific moment in time. Before each instance of data persistence or use, the data should be (re)verified to ensure it is fit for its intended purpose at that specific point in time. We emphasize the critical temporal nature of zero trust in data quality (ZTDQ), highlighting that trust must be continuously reassessed as the data, its usage, and the criticality of the usage can all change. Most importantly, we present a practical and feasible approach for implementing ZTDQ without compromising performance.

Drawing on insights from the cyber security domain, we present this Zero Trust Data Quality (ZTDQ) framework, describe how it replaces traditional reactive and proactive data quality validation, and explain how to:

1. Make it feasible for real time data consumers to verify data quality quickly before every data usage. ZTDQ accomplishes this by pre-checking data quality, and saving the results in tokens that data consumers can quickly check.
2. Make it feasible for data consumers to verify that data is fit for their bespoke purposes at a specific moment in time, even if the data is not fit for other purposes. ZTDQ enforces this through consumer-lens validation of data tokens, ensuring they meet context-specific quality criteria before use.
3. Make it feasible to maximize the value of data quality checks that can be done in a given time window, e.g., before the morning service-level agreement (SLA) deadline of the batch data pipeline. ZTDQ does this by leveraging AI explainability weights to determine the percentage of limited time to invest validating each data attribute.

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[3] The "three lines of defense model" was developed by the Institute of Internal Auditors (IIA). First formally presented in a 2013 IIA position paper titled: *The Three Lines of Defense in Effective Risk Management and Control.* Revised and expanded in 2020 as *"The Three Lines Model."*

[4] Mohammad, M., Talburt, J.R., et. al. (2021). *A Zero Trust Model Based Framework for Data Quality Assessment.* 2021 International Conference on Computational Science and Computational Intelligence (CSCI).

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Goal of Shared Integrity

The rapid advancement of artificial intelligence (AI) and increasingly decentralized data ecosystems is driving a fundamental transformation in how data quality is defined, assured, and maintained. Traditionally treated as a largely static set of validations—checking for completeness, accuracy, or timeliness—data quality management is evolving into a dynamic, continuous process for which assurance must be maintained in real time as data flows across diverse systems, users, and applications. This transformation necessitates a rethinking of trust—not as an assumed attribute, but as an actively defined, verifiable condition embedded into the fabric of data operations and data governance.

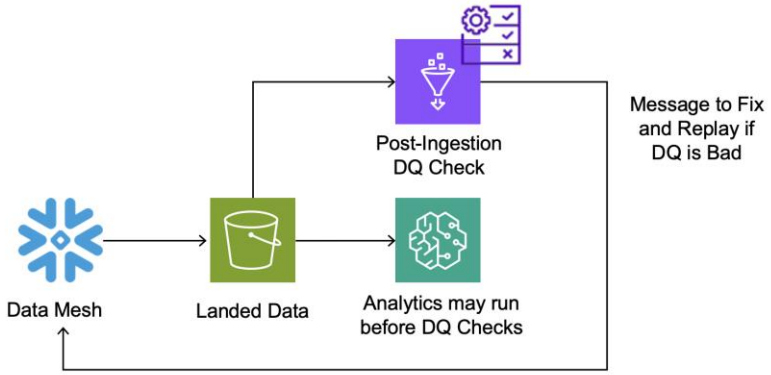
Fundamental in this shift is the Zero Trust security model. Whose core principle— "never trust, always verify"—disrupts the concept of implicit trust within data pipelines. Unlike perimeter-based models that presume internal actors or systems are trustworthy, Zero Trust requires continuous authentication, authorization, and policy validation for every interaction with data, regardless of origin or context. When applied to data quality, Zero Trust reframes what it means for data to be judged reliable, requiring that trustworthiness must be derived from a verifiable security posture, with lineage transparency, access context, and policy adherence.

In this paradigm, Zero Trust is not viewed as a constraint on data usage; it is a catalyst for redefining how data quality is conceptualized and enforced. By embedding Zero Trust principles into the architecture of data quality systems and data management, companies can create continuous assurance loops that monitor and maintain both analytic integrity and security compliance. Ultimately, this integration supports the broader goals of data governance and governable AI—where the quality of data is no longer a compliance checkbox but an active, enforced, and continuously audited component of trustworthy outcomes.

FROM STATIC TO DYNAMIC: THE NEW DEMANDS ON DATA QUALITY

As artificial intelligence (AI), real-time analytics, and distributed computing redefine data-driven operations, conventional approaches to data quality are proving increasingly inadequate. Historically, data quality has centered on periodic validation exercises, guided by established dimensions such as accuracy, completeness, timeliness, and consistency. While these dimensions remain foundational, they were designed for static, centralized, and batch-oriented environments—contexts in which data was largely immutable after ingestion and subjected to limited downstream reuse. Figure One illustrates the traditional reactive data quality validation model.

Figure 1: Traditional Reactive Data Quality Validation



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Due to data delivery SLAs, the most common traditional data validation process allows data consumers to use data as soon as it is prepared, rather than waiting for comprehensive data quality checks to complete. Thus, data quality issues can potentially impact business processes or outcomes. Some such reactive data quality validation systems, e.g., compliance systems for Anti Money Laundering, Trade Surveillance, and Disclosure of Interest, can often replay the pipeline after correcting the data and regenerating their reports with minimal impact beyond the effort devoted to the replay. However, decision making systems, e.g., check cashing, mortgage application, and client onboarding often struggle with reversing a decision made with bad data.

The primary duty of the traditional process is anomaly detection and immediate response: to identify and remediate known data problems as fast as possible. Key steps in this process include:

1. **Issue Detection (Post-Event Identification)**
    a. Data quality issues are often discovered reactively—through user complaints, failed reports, regulatory audits, or anomalies in downstream systems
2. **Root Cause Analysis**
    a. Data engineers begin investigating where the data issue originated and what caused it
3. **Data Profiling and Assessment**
    a. Tools or scripts are used to examine patterns in the data, frequency of errors, and the affected records based on data quality attributes
4. **Rule-Based Validation**
    a. Existing (often manually created) business rules are applied to detect errors conditions in the data (example: format errors)
5. **Manual Remediation or Cleaning**
    a. Depending on the issue: manual correction, or data is reprocessed through modified ETL pipelines
6. **Reporting and Documentation**
    a. Errors, root causes, and fixes are documented.
7. **Patch or Rule Update**
    a. A new validation rule might be created to detect the issue earlier next time.

In contrast, systems without Service Level Agreement (SLA) concerns validate data proactively at the perimeter, before consumers are allowed to use it. This proactive validation detects data problems before they can affect downstream applications or data-driven decisions. The core design principles of proactive data validation include automated rule enforcement, continuous anomaly detection, and seamless integration with data pipeline orchestration frameworks, to ensure data quality is monitored and enforced in real time. For example, in cloud data warehouse environments, proactive data validation may include automated rule enforcement using tools like **dbt** tests (data build tool), continuous anomaly detection through integrated monitoring platforms, and seamless integration through vendor-defined orchestration frameworks. These practices ensure that data quality issues are identified and addressed within the pipeline before the data becomes available for downstream analytics or consumption.

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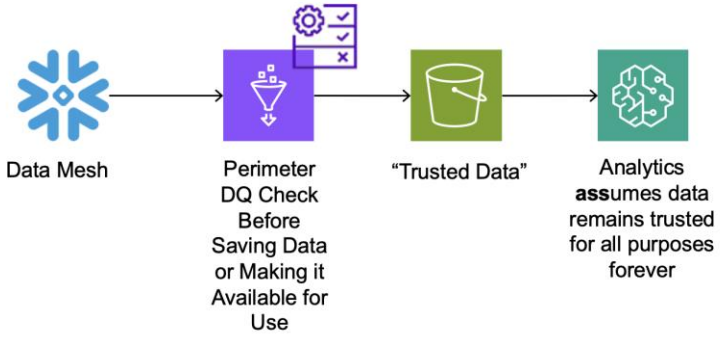
Figure 2: Traditional Proactive Data Quality Validation



Key steps in this validation process include:

1. **Data Quality Framework Definition**
    a. Establish data quality dimensions and define SLAs and KPIs for data quality across domains
2. **Embedded Quality Rules**
    a. Implement validation rules directly into data pipelines (ETL/ELT) or data platforms
    b. Rules are aligned with business logic and applied automatically as data flows
3. **Continuous Data Monitoring**
    a. Use automated tools to monitor data in real-time or near real-time
    b. Alerts are triggered when anomalies or rule violations are detected
4. **Anomaly Detection with ML/AI**
    a. Statistical models and machine learning used to detect outliers or unexpected patterns
5. **Early Warning and Alerting**
    a. Issues are flagged before they impact reports, dashboards, or machine learning models
6. **Incident Management and Root Cause Automation**
    a. Integrated workflows for triaging and resolving issues
    b. Some platforms can auto-identify likely root causes (e.g., upstream table changes, schema drift).
7. **Continuous Feedback & Improvement**
    a. Learnings from incidents are fed back into the system.

Although the proactive approach dramatically improves data quality management over the reactive approach, one major problem exists in both approaches. After data passes quality checks and is classified as trusted, it remains trusted forever no matter what happens to the data itself. If the data is subsequently corrupted, edited, altered, or expired, it still remains trusted. The data also remains trusted forever no matter what happens to the data consumer. Even if the data consumer is an autonomous AI agent capable of performing vast volumes of work without human intervention, the data remains trusted.

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AI'S CHALLENGE AND POTENTIAL TO RE-ARCHITECT THE TRADITIONAL DATA QUALITY MATURITY CURVE

Traditional reactive and proactive data validation models were introduced over thirty years ago. Reactive data validation originated in the 1990s in foundational data models introduced in data warehousing and business intelligence (BI) data engineering. Proactive validation modeling emerged as a formal goal in the early 2000s, influenced by first-generation data engineers like Thomas Redman and Larry English, and data governance frameworks developed by the Data Management Association (DAMA).

However, today's data landscapes are dynamic, decentralized, and increasingly autonomous. Modern data pipelines traverse heterogeneous systems, geopolitical boundaries, and varying trust domains. Data is continuously transformed in transit, consumed by APIs, orchestrated across cloud-native architectures, and operationalized by AI models— often in deployment scenarios unaccounted for during data origination. Under these conditions, Data Quality (DQ) must be assured not only at the point of ingestion, but persistently monitored and validated across all data lifecycle stages. A single corrupted or non-compliant data element can propagate across systems, undermining model performance, data governance, and regulatory posture.

This challenge is especially pronounced in AI and machine learning workflows, where model reliability and fairness are directly contingent on the integrity of training and inference data. In this context, data must not only exhibit statistical validity, but also meet criteria for security, provenance, explainability, and compliance with evolving policy frameworks. Consequently, data quality must evolve into a continuous, context-aware, and risk-sensitive discipline—adaptive to real-time conditions and capable of operating in environments where trust is not assumed but must be explicitly and continually verified.

STATISTICAL VALIDITY VERSUS NON-PARAMETRIC DATA "OBSERVABILITY"

Despite the decades-long evolution and success of BI and data warehousing practices, AI is prompting both deep concern and excitement among data engineers for numerous reasons. Key areas where AI will disrupt current data validation practices include:

AI Doesn't Need "Perfect" Data — But Traditional BI Does

"Traditional business intelligence (BI) systems and data warehouses rely on structured, cleaned, and validated data. In contrast, AI models—particularly large language models (LLMs)—can operate on messy or semi-structured data, often constructing meaning even from incomplete or inconsistent inputs. This has led some optimists to believe that AI can infer itself around the need for data quality validation. AI systems that ignore data quality issues risk generating silent errors—mistakes that go undetected but can undermine the reliability, trustworthiness, and viability of AI-driven solutions.

AI Can Train Itself to Validate Data Without Explicit Rules

Traditional validation uses static rules: e.g., "email must contain @", "age can't be negative". AI anomaly detection models can learn patterns from historical data and flag exceptions without pre-specified rules. Should rule-based data quality verification be replaced by adaptive, ML-driven agents?

AI Agents Can Automate the Entire Data Quality Workflow

Emerging tools—such as AI-powered data observability platforms—leverage LLMs and machine learning models to detect anomalies in data pipelines, recommend or even apply fixes, and automatically generate data quality tests. This raises critical questions: should AI eliminate human data quality roles, such as data stewards, data engineers, or QA analysts? If not eliminate, commoditize their contributions?

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AI Challenges the "Single Version of Truth" Philosophy
Business intelligence and data warehousing were built on the principle of curated, well-governed data—one version of the truth, one schema, one standardized process. In contrast, AI and large language models (LLMs) are designed to process and interpret data from many conflicting, unstructured, or external sources—whether or not those results are accurate or reliable.

TWO NON-PARAMETRIC DATA QUALITY MODELS DO NOT MAKE ONE PARAMETRIC ONE

Traditional data quality validation methods have relied on parametric statistical models designed for detecting anomalies, identifying outliers, and enforcing rule-based consistency checks. In contrast, large language models (LLMs) and other non-parametric machine learning methods are increasingly influencing how data quality validation is addressed in practice. This presents a growing conundrum for data quality professionals: how should AI-driven, non-parametric techniques be integrated into established, statistically grounded frameworks? Will these models augment traditional approaches—or challenge and displace them? A review of current industry trends and expert commentary suggests three areas of convergence, highlighting the differences in approach:

Semantic Pattern Recognition
LLMs can be used to recognize surface-level patterns, such as field format errors, schema mismatches, and obvious mismatches ("Country is US, currency is Yen"). LLM models do this by coding prior data patterns observed in training datasets, not from statistically evaluating actual data distributions.

Prompt-Driven Rule Generation
LLMs can generate validation rules based on schema, metadata, or examples. But they cannot, unless specifically trained on additional contextual factors such as statistical data distributions, business logic, or operational context, validate the correctness of those rules.

Metadata Inspection and Data Commentaries
LLMs can scan metadata, describe table anomalies, or compare against documentation. But these procedures are not performing statistical QA — they are making semantic observations on the data.

THE EVOLVING DATA QUALITY INDUSTRY PARADIGM

The evolving consensus across industry is that data quality validation in practice can no longer rely on a single method or toolset, but instead requires an integrated, multi-layered approach. The evolving practice in use is grounded in three core modeling paradigms:

- Semantic models (e.g., LLMs) offer flexible, context-aware interpretations of data, enabling data-driven rule suggestions and metadata alignment.
- Statistical models provide the analytical rigor needed for detecting anomalies, data drift, and distributional inconsistencies.
- Rule-based systems enforce deterministic, business-defined constraints that ensure structural and logical consistency.

Individually, each approach has its own strengths and limitations. Taken together, they can form a complementary framework balancing contextual interpretability, statistical precision, and rule-based data governance.

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UNIFYING DATA QUALITY AND CYBER SECURITY WITH ZERO TRUST

The traditional separation between data quality and data security is converging in modern data architectures. As organizations adopt cloud-native platforms, real-time data pipelines, and AI-driven analytics, the attack surface for both data integrity and misuse expands significantly. In this context, Zero Trust—a security model predicated on continuous verification, least-privilege access, and the assumption of breach—offers a valuable conceptual framework for reexamining how data quality is governed and enforced.[5]

While framed around user authentication and endpoint access, the implications of cybersecurity's Zero Trust principle of "never trust, always verify" can be extended beyond identities and infrastructure to encompass data verification itself.[6] Applied to data, Zero Trust re-conceptualizes validation as a dynamic, enforceable, and auditable state rather than a static control. This implies that every data interaction—whether read, write, transformation, or inference—must be explicitly authorized, verifiable in real time, and immutably logged for accountability and forensic traceability.

Operationalizing Zero Trust within data quality workflows requires a shift from periodic, rule-based validation to continuous, contextual, and policy-driven assurance. This involves embedding verification logic not only at the point of ingestion, but persistently across the data lifecycle—spanning transformation pipelines, storage environments, and analytic endpoints. Critical mechanisms include provenance tracking, real-time anomaly detection, schema enforcement, and behavioral profiling of data flows. In parallel, metadata management and data contracts serve as operational interfaces for defining and enforcing data quality expectations between producers and consumers. By treating data as an active and verifiable asset, rather than a passive artifact, organizations can architect a more resilient and adaptive trust posture across their data infrastructure.

Implementing ZT-Aligned Data Quality

To implement this model effectively, organizations must adopt architectural patterns that support declarative trust policies, automated validation checkpoints, and policy-aware data lineage tracing. Platforms must be capable of introspecting data flows, validating assertions dynamically, and escalating violations through governance or security controls. Integrating data quality enforcement with identity, access, and policy engines allows for end-to-end traceability and response-blurring the lines between observability, quality assurance, and threat detection. Ultimately, Zero Trust–aligned data quality is not merely a convergence of two disciplines, but an underlying practice for governing trustworthy, automated, and scalable data ecosystems in increasingly adversarial and compliance-driven environments. Table 1 summarizes these points.

[5] Newman, Lily Hay (2021). *What Is Zero Trust? It Depends What You Want to Hear: The cybersecurity world's favorite catchphrase isn't any one product or system, but a holistic approach to minimizing damage.* Wired, September 12, 2021. Available at: https://www.wired.com/story/what-is-zero-trust/

[6] Knehr, Will (2025). *Zero trust: The next evolution in cybersecurity*. Security Magazine. Available at: https://www.securitymagazine.com/articles/101074-zero-trust-the-next-evolution-in-cybersecurity?

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Table 1: Zero Trust–Aligned Data Quality: A Progressive Policy Framework

Policy Stage	Strategic Objective	Key Principle	Implication for Data Quality
1. Converge Data Quality & Security	Acknowledge that data quality and security are interdependent in modern architectures.	Integrity and protection should both be managed with zero trust and potentially leverage the same components.	Treat data integrity the same way as security; integrate assurance into platform strategy.
2. Extend Zero Trust to Data	Apply "never trust, always verify" to data interactions, not just users and endpoints.	Trust must be continuously verified and enforced.	Require validation of every data read, write, or transformation; embed auditability.
3. Implement Lifecycle Verification	Shift from static, periodic testing to dynamic, policy-driven quality enforcement.	Data must be validated in context, continuously.	Embed checks at ingestion, transformation, and delivery; monitor behavior over time.
4. Operationalize with Policy Controls	Use platform-integrated policies to enforce quality and traceability at scale.	Quality and access policies must be declarative and enforceable.	Connect data validation to identity, access, and governance engines; enforce end-to-end lineage.

The framework presented here defines ZTDQ as a sequence of data governance decisions, capturing how companies could operationalize Zero Trust principles within data quality management. As stated earlier, in distributed, federated data environments, trust cannot be based solely on static metadata about data origin or assumptions about the integrity of upstream systems and custodianship. These signals may be far removed from the point of data creation or the repurposing of data. In this light, Zero Trust can be seen to evolve from a security posture into an essential design paradigm for data infrastructure—one in which data quality, trust, and security are not parallel concerns but interdependent system properties that are assured continuously by design.

A PRACTIONIONER'S GUIDE TO IMPLEMENTING ZERO TRUST PRINCIPLES INTO DATA QUALITY

To operationalize the synergy between Zero Trust architectures and next-generation data quality, organizations must move beyond periodic data quality checks performed only at isolated points in the data lifecycle—such as production, aggregation, or consumption. Traditional dimensions of data quality—accuracy, completeness, consistency, integrity, and others-remain essential, but must now be evaluated continuously and contextually. A key innovation in this model is the concept of a 'data quality token': a verifiable artifact that captures the results of quality checks performed at the point of data persistence. Before any data is stored, a token is generated to represent its validated quality state; before any data is consumed, that token is verified to ensure that the data meets predefined trust thresholds. This creates a continuous, enforceable chain of data trust. Moreover, every check should be auditable, creating a transparent, traceable log of data quality assurance activities that align with Zero Trust principles of verification before access.

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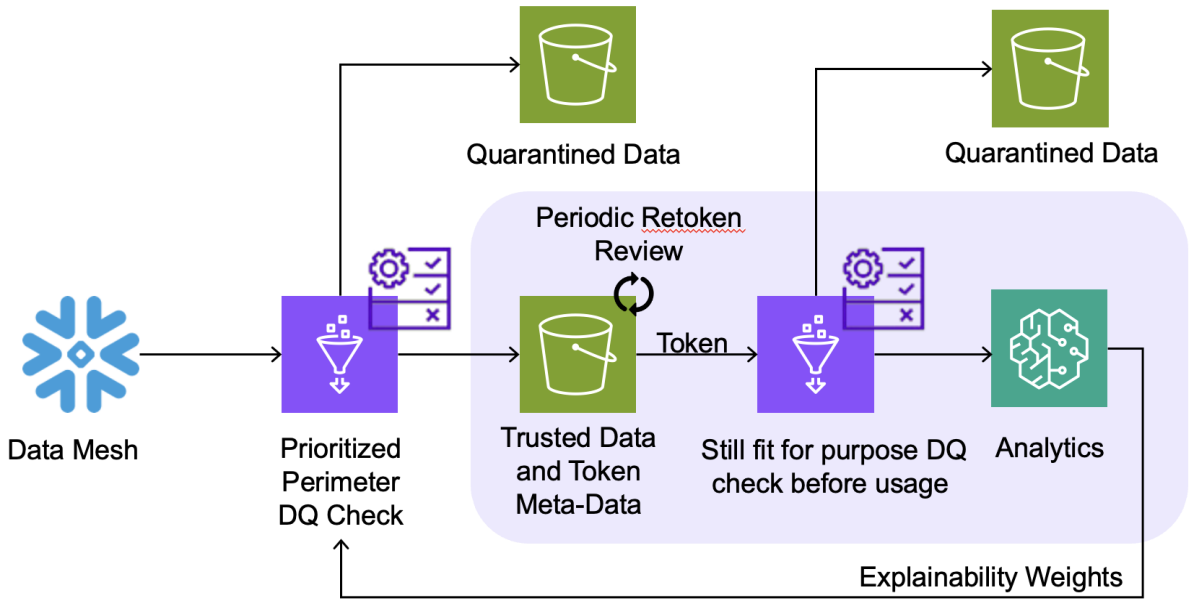
Beyond evaluating traditional data quality dimensions, organizations should also monitor operational metrics—such as the time required to pre-validate data and generate or verify data quality tokens—as part of system performance and observability. Additionally, to strengthen trust in AI-driven processes, companies should continuously log the global explainability weights of machine learning models. These weights, which reflect the relative influence of features on model outputs, should be incorporated into perimeter data quality checks to ensure that downstream data use aligns with transparency and accountability standards. These practices enable organizations to build continuous monitoring systems that treat data quality as a dynamic condition—one shaped not only by intrinsic semantic properties but also by the context of data access, transformation, and usage. Figure 3 illustrates this process.

Figure 3: Implementing Zero Trust Principles into Data Quality



Translating Zero Trust principles into operational data quality requires a deliberate architectural approach—one that integrates quality monitoring into a unified, real-time ecosystem.[7]

Following are the key architectural design components in implementing Zero Trust principles:

[7] IBM Blog (2023). *The Evolution of Zero Trust and the Frameworks that Guide It*. April 18, 2023. Available at: https://www.ibm.com/think/insights/the-evolution-of-zero-trust-and-the-frameworks-that-guide-it

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Component	Type	Description
Data Distributions	Data Strategy Best Practice	Defines the attributes that a producer will provide, the physical method of distribution (File, API, Kafka, etc.), and the schedule for distribution
Data Contracts	Data Quality Best Practice	These are the terms of service offered by the data producer which include the delivery time SLAs and data quality checks that will be met by the producer. Most typically, the consumer accepts the terms of service as offered by the producer for all consumers. On an exception basis, the consumer may negotiate to change the general terms of service or to create bespoke terms of service.
Data Contract Consumer Lens	Data Strategy Best Practice	Data consumers may use a subset of the attributes in a distribution or require a subset of the data quality checks in a data contract. By defining a consumer lens, a consumer informs the producer of what the consumer actually consumes and expects so that the producer can notify consumers of just the issues that impact the consumer.
Perimeter Data Quality Pre-Checks	Data Quality Best Practice	Execute all data quality checks defined in the data contract before data enters the system and store the results in a data quality token. Only data that passes the checks should enter the system. Data that fails checks should be quarantined for investigation.
Data Quality Tokens	Novel Contribution	The tokens are meta data that include: ▪ The source of the data (data producer) ▪ The data contract ▪ The results of the perimeter data quality checks ▪ A checksum that a consumer can review to ensure that the data has not changed since it was validated. The tokens must be created at the same granularity in which the data will be used. Often this means that there should be one token per row, e.g., one token per transaction, on per account, etc.
Data Quality Token Checks	Novel Contribution	Before every data usage, data consumers should check the token to ensure that the quality is currently fit for the particular purpose. If the same data is used repeatedly for either the same or different purposes, the token should be checked repeatedly before every usage.

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Periodic Retoken Review	Novel Contribution	Periodically validate that the historical tokens will be accepted by the current AI analytics applications. For example, the token must be regenerated if it contains the results of data validation against an old data contract but the AI requires data quality to the level of a newer data contract version.
Global Explainability Weights	AI Best Practice	Generate global explainability weights for all machine learning models with an algorithm such as SHAP.
Quality Check Prioritization	Novel Contribution	Feed the global explainability weights into the perimeter data quality check process. In the commonplace event that there is not time to complete all data quality checks on all attributes for all data rows, ensure that checks receive attention in proportion to their importance. For example, if an attribute's explainability weight accounts for 11% of a model's decision, then spend 11% of the available time checking that attribute's quality.

By aligning data quality controls with Zero Trust principles, organizations can implement a model in which data quality and trustworthiness are treated as continuously verifiable properties—evaluated at each point of access and enforced through policy-driven mechanisms. This ensures that AI systems and analytical processes operate exclusively on data that meets defined quality standards.

CHALLENGES AND FUTURE DIRECTIONS IN IMPLEMENTING THE NEW PARADIGM

While the integration of Zero Trust principles into data quality systems offers a compelling framework for governable, AI-ready data, it also introduces an initial uplift to build the components listed in the previous section as well as effort to maintain them. This effort is in addition to current efforts and thus must be acknowledged and addressed for the paradigm to be widely adopted and effectively implemented. We consider the effort necessary to properly support AI. The effort is also potentially offset by the reduced operational effort of reacting to data quality issues and cleaning up bad decisions and bad outcomes due to data quality issues. We turn now to the challenges – integration complexity, organizational silos, the resistance to change legacy systems, and the need to define and standardize new metrics.

IMPLEMENTATION CHALLENGES

1. Integration Complexity

Implementing Zero Trust data quality requires tightly coupling systems that have traditionally operated in silos—security platforms, data pipelines, metadata systems, and quality monitoring tools. This integration involves:

- Custom middleware or APIs to pass context (e.g., identity, risk scores) between systems.
- Upgrades to legacy infrastructure to support real-time verification and logging.
- A new class of telemetry: capturing not just data content but data behavior and access conditions.
- Without a common architecture or interoperability standards, the cost and complexity of implementation may slow adoption.

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2. Organizational Silos

Security teams and data teams often operate under different priorities and governance models. Security emphasizes control and restriction; data teams prioritize access, availability, and velocity. Embedding Zero Trust into data quality requires:

- Shared governance frameworks and joint accountability models.
- Cross-functional fluency: security teams must understand DQ implications, and vice versa.
- Cultural shifts in how “trust” is defined and enforced across the organization.
- Bridging this divide requires not only tools but also shared incentives and executive sponsorship.

3. Resistance from Legacy Systems

Many existing systems are not designed for dynamic policy enforcement or fine-grained access telemetry. Challenges include:

- Limited ability to instrument legacy data stores for real-time access logging.
- Inflexible data models that can't accommodate enriched metadata or policy hooks.
- Lack of auditability or lineage support in older ETL tools and batch systems.
- Retrofitting Zero Trust principles into legacy architectures may require phased approaches or parallel modernization efforts.

4. Defining and Standardizing New Metrics

Metrics like "policy conformance," "access risk," or "trust score" are still emerging and lack formal industry definitions. This complicates:

- Benchmarking and interoperability across organizations.
- Regulatory acceptance of Zero Trust-aligned data quality frameworks.
- Automation of validation and scoring across diverse ecosystems.
- The field lacks a shared language for Zero Trust–aligned data quality metrics, which hinders measurement, validation, and trust.

FUTURE DIRECTIONS

Despite these challenges, several emerging trends suggest a path forward—and even a potential transformation of the data quality discipline itself.

1. AI-Driven Trust and Quality Scoring

Machine learning models can analyze patterns of data access, transformation, and usage to:

- Detect anomalous behavior affecting data quality.
- Predict trustworthiness of new datasets based on historical patterns.
- Automate quality scoring by combining semantic, behavioral, and security dimensions.
- AI becomes not just a consumer of data quality, but a contributor to its dynamic enforcement.

2. Formalization of Zero Trust Data Contracts

As data sharing becomes more frequent across organizational and geographic boundaries, we will likely see the emergence of:

- Policy-enforced data contracts that specify not only schema but also permitted access, usage, and retention conditions.
- Runtime-enforced contracts embedded into APIs and pipelines to prevent non-compliant usage in real time.
- Data contracts will evolve from descriptive agreements to executable, enforceable policies—creating self-governing data ecosystems.

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3. Standardization and Framework Development

Industry consortia, regulatory bodies, and open-source communities are beginning to explore:

- Common frameworks for Zero Trust data architecture (e.g., NIST ZT Data Pillar guidance).
- Interoperability standards for lineage, access metadata, and audit logging.
- Certifications or benchmarks for AI systems trained on Zero Trust–compliant data.
- A new layer of infrastructure—akin to DevSecOps—may emerge around DataSecQualOps: the integration of data security, quality, and operations.

4. Alignment with Responsible and Governable AI

As regulatory and public pressure for explainable, accountable AI grows, the demand for trusted data inputs will intensify. Zero Trust data quality aligns directly with:

- Model transparency: Traceable data inputs support post-hoc explanation and validation.
- Bias mitigation: Secure lineage enables better understanding of dataset composition and risks.
- Audit readiness: Continuous assurance of data access and transformation supports compliance with AI accountability frameworks.

CONCLUDING REMARKS: TOWARD TRUSTWORTHY AI THROUGH ZTDQ

As companies increasingly rely on hard-to-explain and autonomous AI to drive decisions, the fundamental meaning of DQ is shifting. No longer a passive or static attribute, DQ must now be continuously assessed, contextually validated, and actively enforced. In this paradigm, trust in data is not an assumption—but a condition that must be earned, verified, and preserved throughout the data lifecycle.

Zero Trust provides a resilient foundation for enhancing DQ across modern, distributed environments.[8] Combined with active data quality, Zero Trust offers a new foundation for data assurance. When embedded into data platforms and analytic workflows, Zero Trust transforms data quality from a technical checkpoint into a data governance mechanism that ensures that data consumed is fit for the bespoke purpose of the moment and does not cause silent errors in autonomous agentic AI systems.

This convergence enables what we define as Zero Trust Data Quality (ZTDQ): a model where data is verified not just at the perimeter, but before every use. Data is deemed "high quality" only when the consumer reviews and accepts it immediately prior to use. To make ZTDQ performant, we adopt a security-inspired pre-checking approach—data is checked once, and the result is stored in a token used for subsequent verifications. To optimize time-constrained checks, we use our model's global explainability weights to prioritize efforts. Crucial to this model are emergent metrics like verifiable lineage, exposure history, policy conformance, and access-context metadata—dimensions often missed by traditional frameworks but essential in AI-driven, regulated environments.

Implementing the ZTDQ vision is not without challenges. It requires an initial uplift to institute data contracts, save data quality tokens, and collect AI explainability weights, and the formalization of new standards and metrics. Yet if companies commit to this transformation, organizations will gain the ability to operate AI and analytics systems that are not only more effective, but also more accountable, auditable, and aligned with changing regulatory and societal expectations for responsible technology deployments, particularly autonomous agentic AI deployments.

[8] White House Executive Order 14028 mandated that all federal agencies adopt the zero trust cyber security model, e.g., DoD for nuclear command and control, and DHS for citizen records and biometrics systems. Zero Trust is explicitly included in the most prevalent cyber security framework: NIST 2.0 CSF.

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Future Research

The authors plan to continue exploring the role of AI explainability weights in advancing data quality, by introducing a framework, methodology, and practical guidelines for driving data quality decision-making in proportion to data attribute importance, as inferred from model behavior. This approach will seek to align data quality investment and validation intensity with the real-world influence of individual attributes on downstream AI outcomes. Important steps in this work will be to map attribute-level quality importance, and continued work on token-based and/or metadata-driven model architecture, among other topics.

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Disclosure

The analysis and views presented in this brief reflect the authors' independent work and are not intended to represent the views of any organization or individual affiliated with their employers, the SPARK AI Consortium, or affiliated organizations.

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About the SPARK AI Consortium

The SPARK AI Consortium was founded in 2024 as a response to the rapidly evolving AI landscape and the growing need for interdisciplinary collaboration to address the complex challenges of AI implementation in business contexts.

Hosted at the San Diego Supercomputer Center (SDSC), SPARK AI brings together the technical expertise of Center AI researchers in advanced computation, data science, and integrative AI solutions with the expertise, experience, and practical insights of industry leaders to create a unique ecosystem for innovation and thought leadership.

The consortium is built around an eclectic mix of AI experts, senior data professionals, general managers, and academic researchers, united by a common goal: to ensure that AI technologies are developed and deployed in ways that create sustainable value for businesses while benefiting society as a whole.

An advisory board made up of corporate members, research faculty and industry experts guide the consortium in its strategy, operations and publication of research outputs.

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